SOFTWARE REQUIREMENT SPECIFICATION DOCUMENT (SRS)

TEAM MEMBERS:

TURAB HASHMI - BSSE23002

LAIBA AHMAD - BSSE23085

MANAL RANA - BSSE23099

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National Accountability Bureau.

1.Intro

1.1 Purpose

This document, Software Requirement Specification (SRS) defines the functional and nonfunctional requirements related to the modifications proposed in the existing web portal belonging to NAB. As part of the procedure, the present solutions provided by the National Accountability Bureau would be integrated under a unified system, i.e., Integrated Accountability System. This system is proposed to serve the purpose of being a centralized hub for communications, collaboration, accountability, and operational efficiencies integrating case and document management systems with performance tracking and collaboration tools all in one facility.

1.2 Scope

The Integrated Accountability System will enhance the existing capabilities of the website by introducing new systems and services from features. Additional enhancement and modification to the existing site are as follows:

Centralized Messaging: Secure messaging and notice features will be enabled so that number of employees can communicate with departments directly using the web portal.

Task and Document Management: Shared tools will be integrated for document management and task assignments for users that will offer the allowance to manage and collaborate on files using the already existing NAB site framework.

Performance Dashboards- Add performance tracking features like KPI monitoring and productivity dashboards into the same website for department heads and managers.

e-training: The online training module shall be made available at website level for user orientation and back office onboarding.

These would be built on top of NAB's existing infrastructure while minimizing disruption and ensuring that it smoothly integrates into the current environment.

1.3 Definitions, Acronyms, and Abbreviations

NAB: National Accountability Bureau

API: Application Programming Interface

KPI: Key Performance Indicator

SRS: Software Requirements Specification

UI/UX: User Interface/User Experience

AWS: Amazon Web Services

Azure: Microsoft Azure

HTTPS: Hypertext Transfer Protocol Secure

1.4 References

IEEE Standard 830-1988, "Recommended Practice for Software Requirements Specifications."

NAB internal policies and procedural guidelines for system security, data integrity, and integration protocols.

The Integrated Accountability System requirements are indeed the new features to be added into NAB's live website. The system centrally organizes all communications and streamlines workflow and improves collaboration with all departments. The document regarding functional and non-functional requirements, system dependencies, and integration considerations so that the project goes through successfully and is seamlessly deployed to NAB's current web platform.

2. General Description

2.1 - Product Perspective

Integrated Accountability System would be an add-on extension to the NAB's currently existing website-in which the user will be performing in a seamless manner while ensuring the sanctity of the present platform. The system is expected to leverage maximum on the existing infrastructure of NAB and integrate into the current case management and document management systems. Accessed through modern web browsers, the web-based application would be securely hosted on cloud platforms in order to achieve scalability and accessibility.

2.2 - Product Features

Centralized Communication: Secure messaging and notifications for employees directly embedded in the user interface of the NAB website.

Integrated Workflow: Seamless task and data sharing through NAB website, via API-integrating existing case and document management systems.

Collaboration Tools: Shared calendars, editing documents, and managing tasks through an advanced interface to the existing NAB website.

Performance Dashboards: New website feature to monitor KPIs, task completion rates and total productivity.

Training Module: E-Resources for new staff onboarding including tutorials, guides and videos but accessible via the NAB website

2.3 -User Classifications and Features

System Administrators: They are the administrators responsible for the functionality of the system without compromising data security and integrity; hence, they have access to backend features and configuration settings of the system.

Department Heads: They monitor progress through performance dashboards and task management tools. They will also see advanced analytical tools for reporting and decision-making.

Employees: Collaboration, communication, and task management tools integrated into the NAB website. They will be able to work together on tasks, track performance, and receive notifications within the system.

2.4 - Operating Environment

Software: This will be a web application that will run on modern browsers like Chrome, Edge, Firefox, and will be fully integrated into NAB's existing website.

Hardware: The application will run on existing desktop and laptop hardware with a minimum requirement of 4GB RAM and dual-core processor.

Hosting Platform: The application will be hosted on a cloud platform such as AWS or Azure, allowing availability (99.9% uptime) as well as scalable system solution.

2.5 - Assumptions and Dependencies

The existing NAB case and document management systems are available through API.

The new system will also be accessible through existing NAB website accounts for the users. There will be reliable internet access for the users for reaching the refurbished NAB website. Employees will have enough computer skills to keep up with using new features on the website with minimum training required.

3 - The provisions for specification

3.1 - Functional requirements

Secure Messaging: The communication system must allow specific users to send and receive messages securely using the NAB website.

New message notifications for users: Tasks, documents updated, and performance metrics must use messages under this system.

Task and Document Management: This system will enable users to create, allocate, and monitor tasks strictly through the website. The system must enable users to load, manage, edit, and collaborate with various stakeholders in real time on the interface of the website.

Performance Dashboards: The system will feature dashboards "stars-on" embedded within the website, which display key performance indicators, for instance, task completion rates and statistics on productivity.

3.2 - Non-functional Requirements

User experience: Not exceeding 2 seconds response time to queries by users when used through the NAB website. Through this system, up to 500 concurrent users should not experience performance degradation.

Usability: The system should be easy to use with no more than two hours of training needed for basic usage. The site will be responsive to all display sizes and all devices; consequently, the user experience remains an authenticated user on the NAB website.

Security: Messaging and all kinds of data transfers within the system must be over HTTPS and supported with end-to-end encryption. User authentication will have a multi-factor authentication (MFA)-based security measure.

Availability: 99.9% uptime for the system except the time of scheduled maintenance for making available to users continuously through NAB website.

Scalability: This organization may extend itself in all dimensions towards future growth snowballing with respect to numbers of users and physical size of data. From 500 concurrent users, the organization should be scalable.

3.3 - External Interface Requirements

User interface: Integrating the user interface into the NAB websites while allowing accessibility through web browsers, via responsive design, to ensure optimal access across devices.
Hardware Interface: The existing desktop and laptop hardware is not to incur further investment as far as software is concerned.
Software Interface: It must integrate into NAB's existing systems through RESTful APIs to provide a secure data-exchange route with highly streamlined functionality.
Communication Interface: The system is using HTTPS for a secure data transmission between user's interface and backend servers.

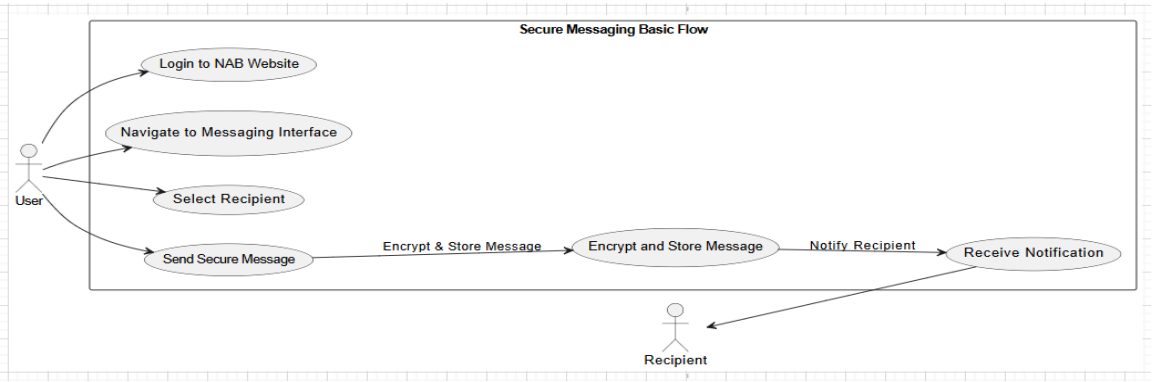
4 - USE CASES

Here are the use cases and alternate use cases based on your provided requirements:

4.1 - Use Case 1: Secure Messaging

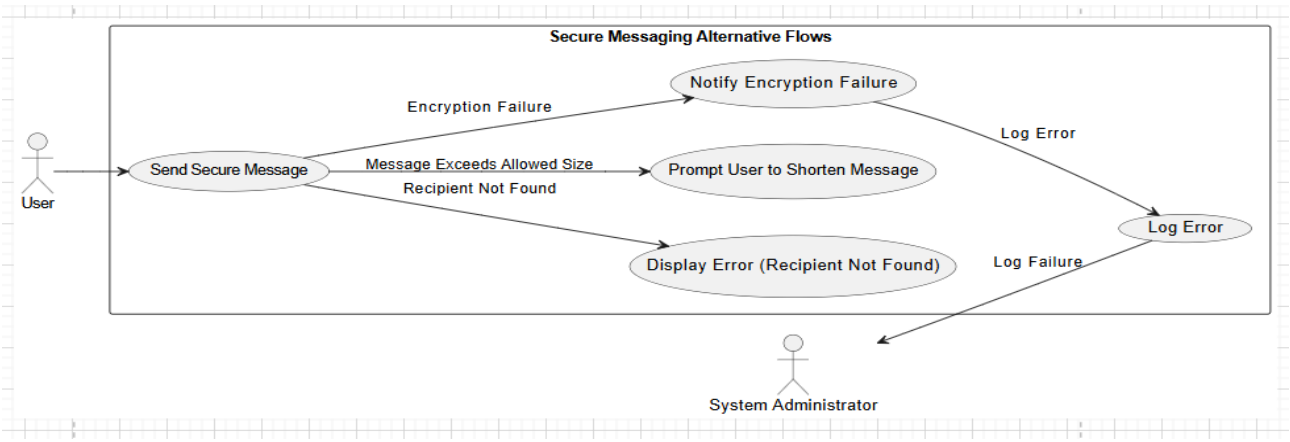
Basic Flow:

1. User logs into the NAB website.
2. User navigates to the messaging interface.
3. User selects the recipient from a list of authorized users.
4. User types and sends a secure message.
5. Recipient receives a notification about the new message.
6. Message is encrypted and stored securely in the system.

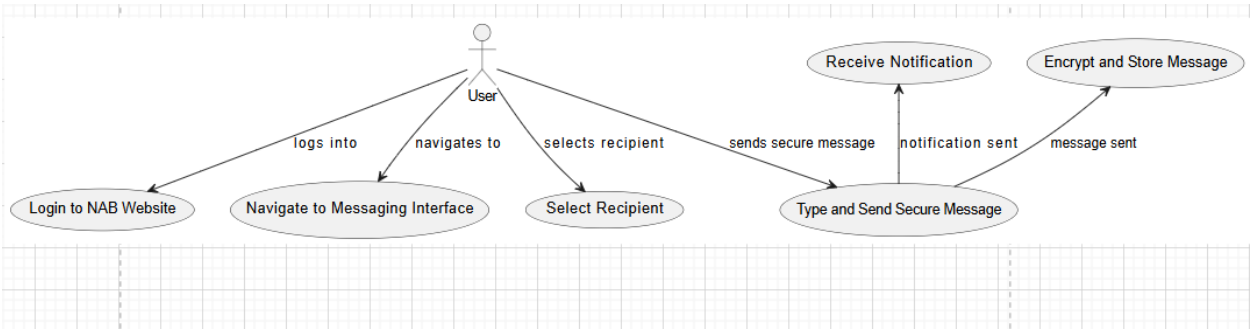


Alternate Flows:

- Recipient not found: The user tries to send a message to a user who is not part of the system. The system displays an error message.
- Message exceeds allowed size: If the message exceeds a predefined size limit, the system prompts the user to shorten it.
- Message encryption failure: The system cannot encrypt the message due to a technical error and notifies the user to try again later.



Test Case:

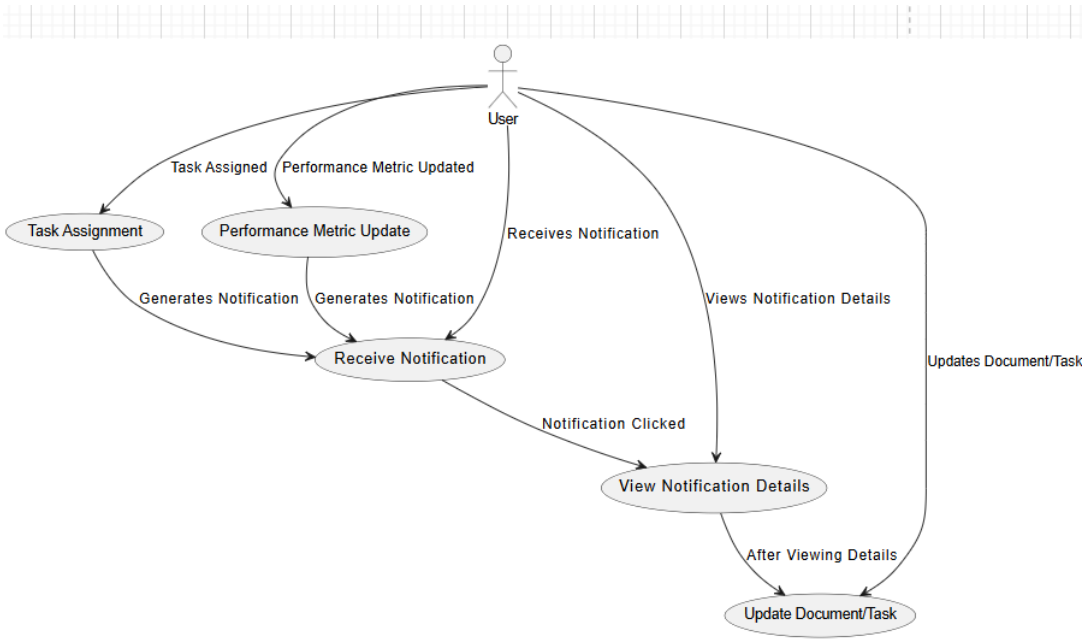


4.2 - Use Case 2: New Message Notifications

Basic Flow:

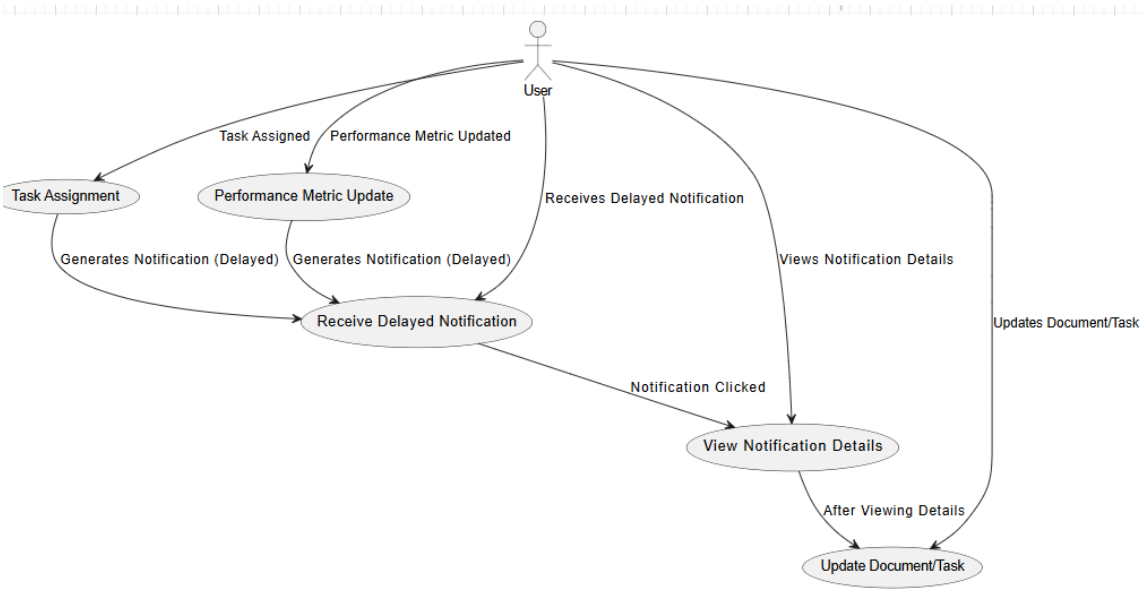
1. **System generates notifications**:
   - When a **task** is assigned to the user.
   - When a **performance metric** has been updated by the system for the user.
2. The **user receives a notification** on the **dashboard** or via **email**.
3. The **user clicks** on the notification to view the details.

4. The **user may update** the document or task after viewing the details.

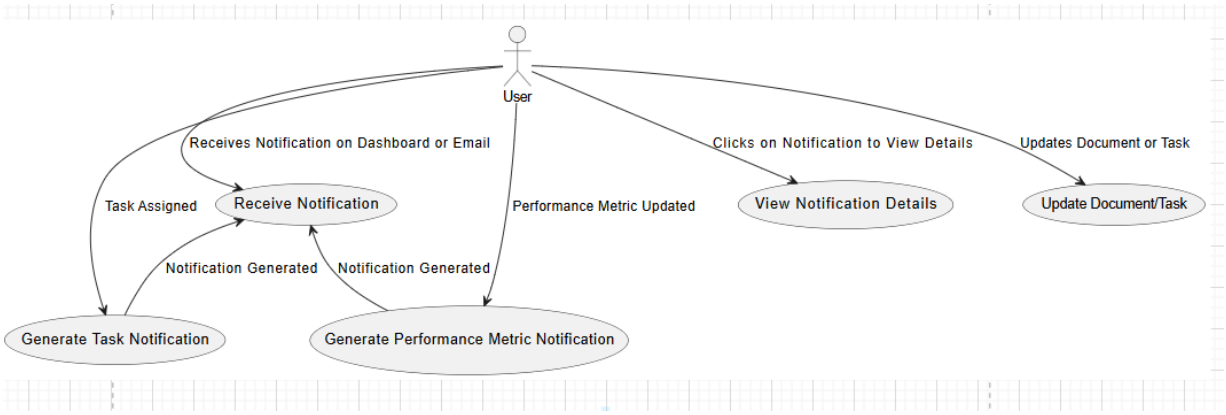


Alternate Flows:

- **Notification Delay**: If the notification system is delayed due to technical reasons, the system queues the notifications and sends them once it's operational.
- **User Disabled Notifications**: If the user has opted out of notifications, the system suppresses the alerts according to the user's preferences.



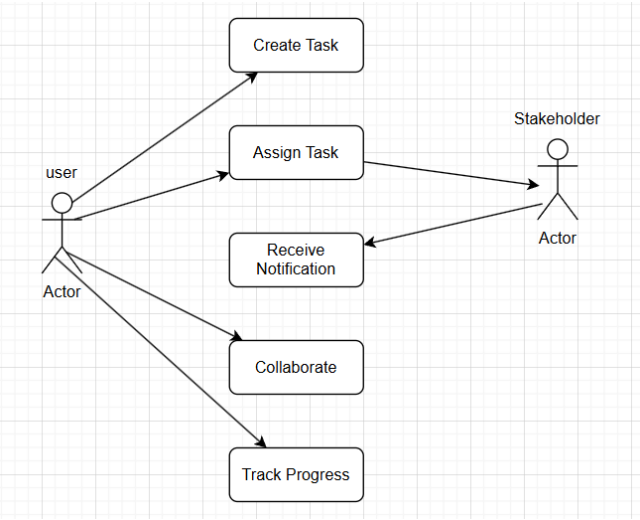
Test case:



4.3 - Use Case 3: Task and Document Management

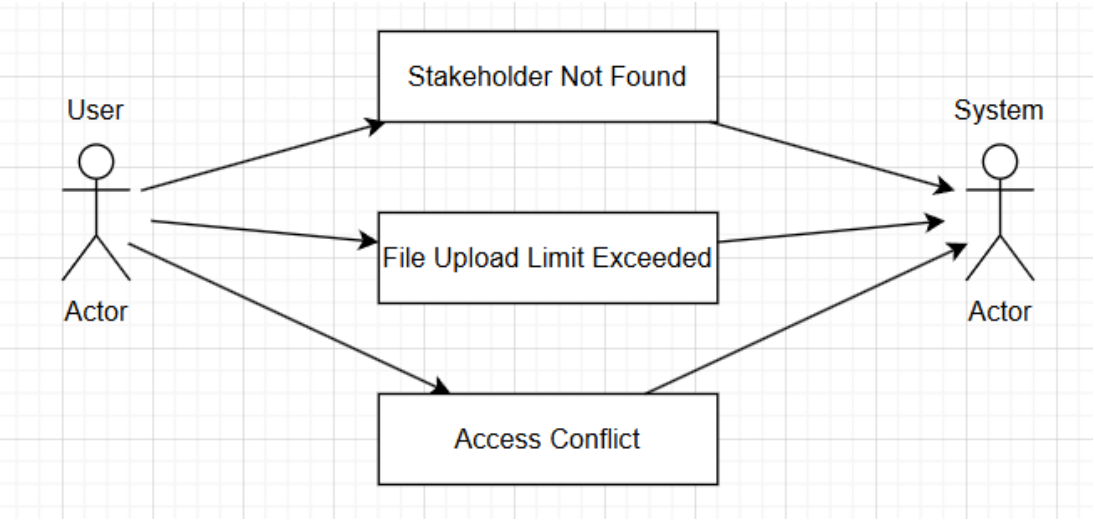
Basic Flow:

1. The **user creates a new task** and provides details such as title, description, and deadline.
2. The **user assigns the task** to one or more stakeholders.
3. The **stakeholders receive notifications** about the task assignment (from Use Case 2).
4. The **user and stakeholders collaborate** in real-time on attached documents using the website interface.
5. The **user tracks the progress** of the task and updates the task details as necessary.

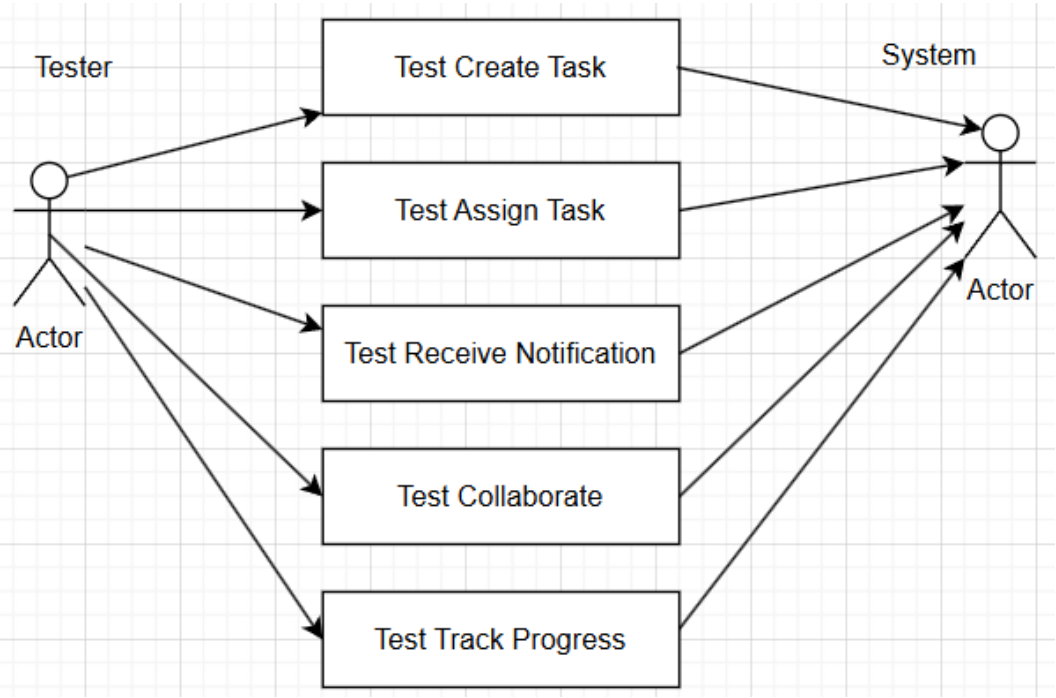


Alternate Flows:

- **Stakeholder Not Found**: If a selected stakeholder is not a registered user, the system displays an error message.
- **File Upload Limit Exceeded**: If the user attempts to upload a document that exceeds size limits, the system prompts the user to compress or resize the file.
- **Access Conflict**: If multiple users try to edit the document simultaneously, the system handles conflict by saving changes sequentially or prompting the users to retry.



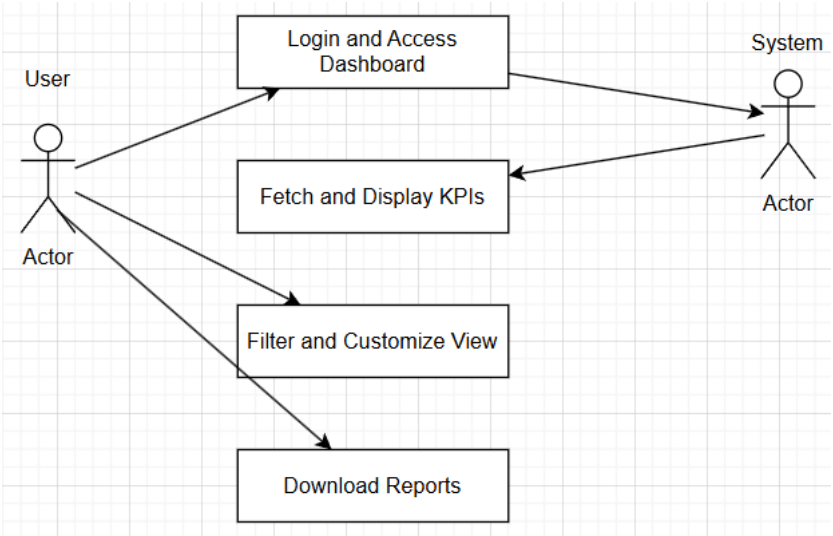
Test case:



4.4 - Use Case 4: Performance Dashboards

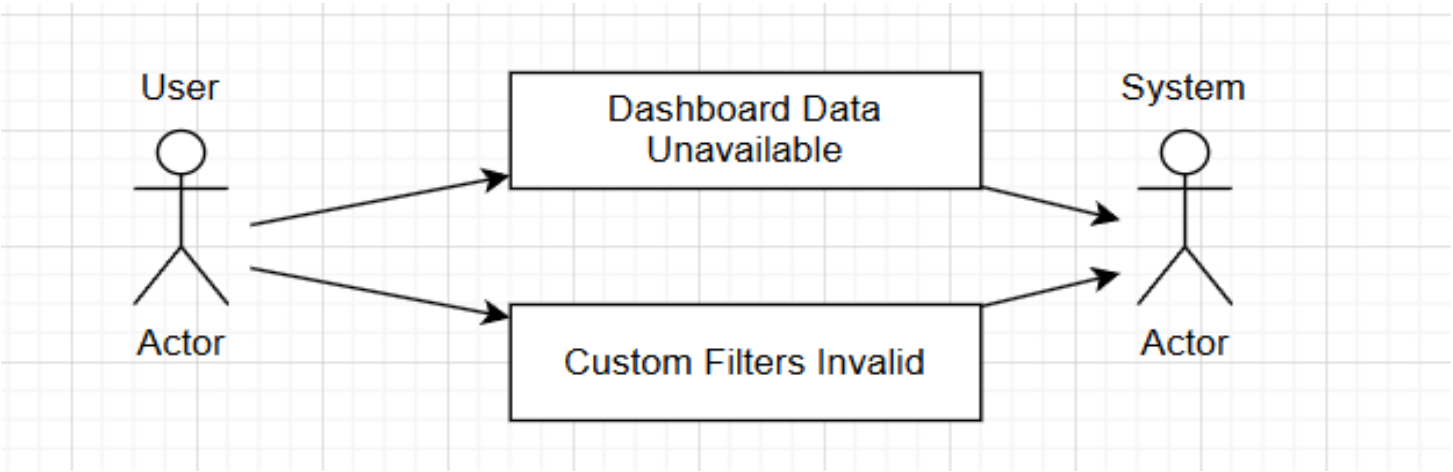
Basic Flow:

1. The user logs into the NAB website and accesses the performance dashboard.
2. The system fetches and displays key performance indicators, such as:
   - Task completion rates.
   - Productivity statistics.
3. The user can filter and customize the dashboard view based on department, time range, or individual/team performance.
4. The user downloads report as needed.

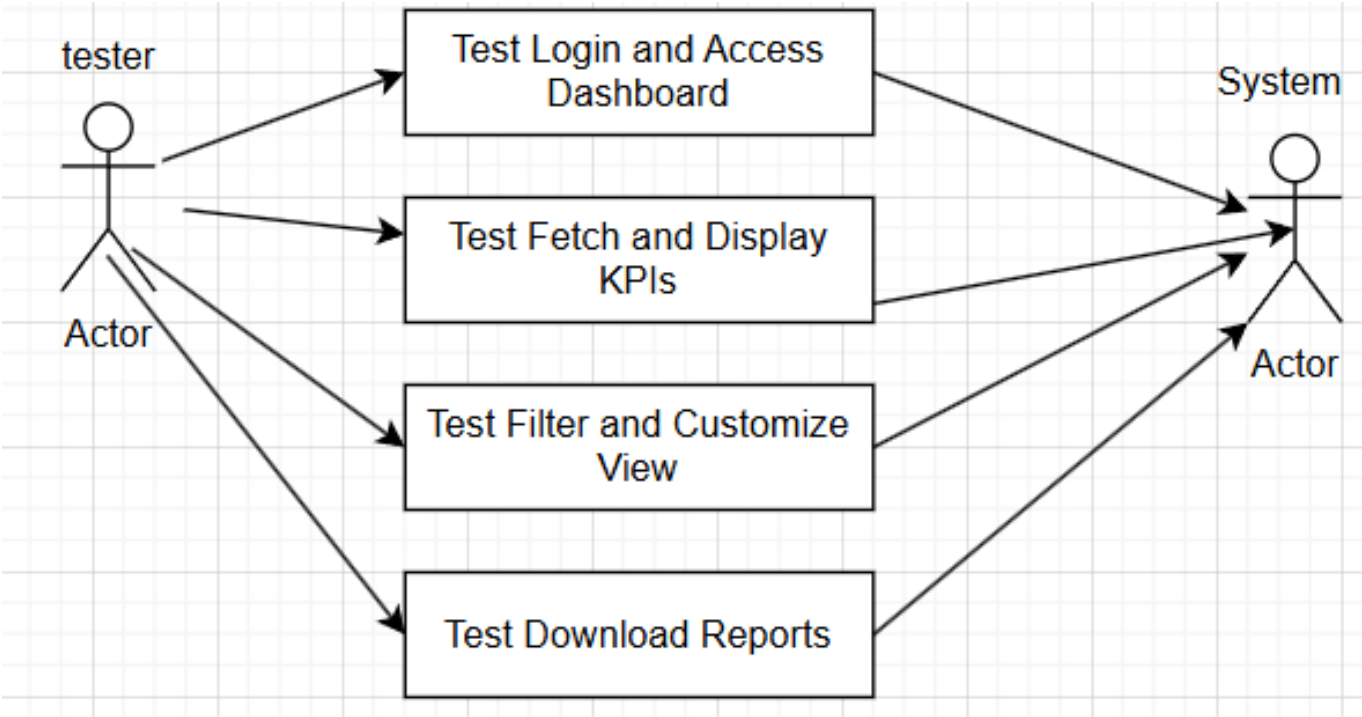


Alternate Flows:

- Dashboard data unavailable: If the data required for KPIs is missing or delayed, the system displays placeholders and prompts the user to check back later.
- Custom filters invalid: If the user applies an invalid filter, the system reverts to default settings and notifies the user.

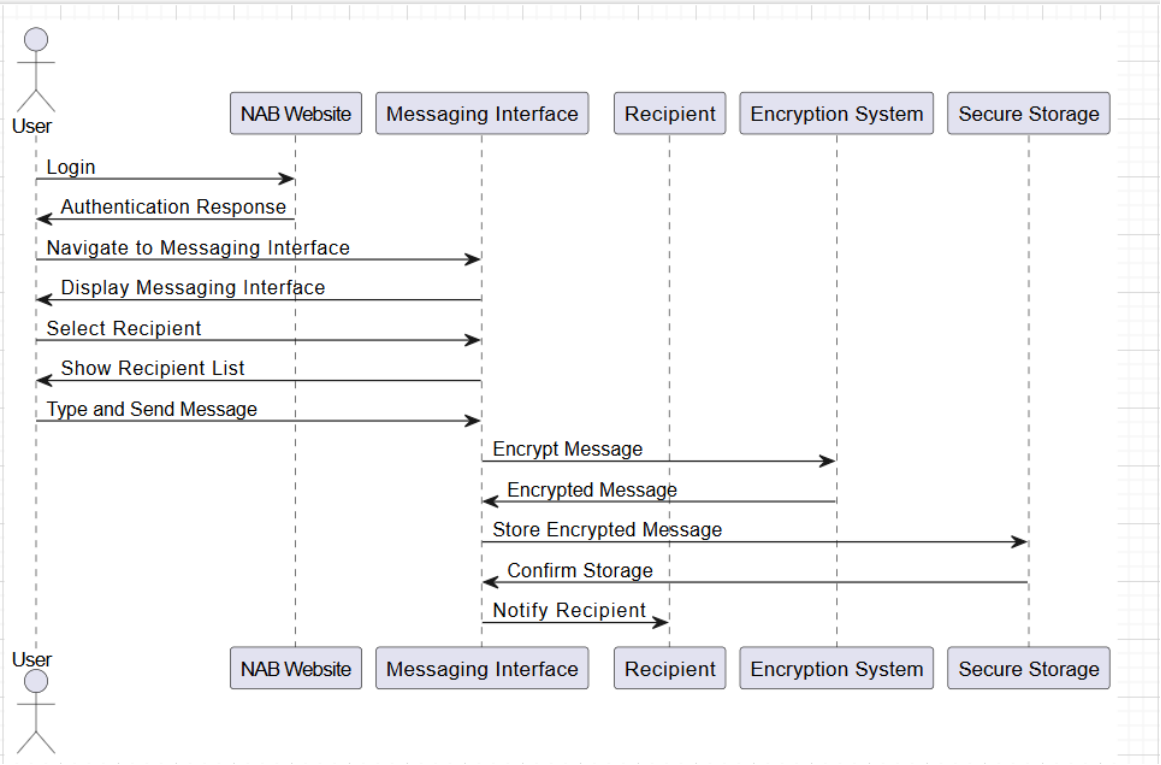


Test case:

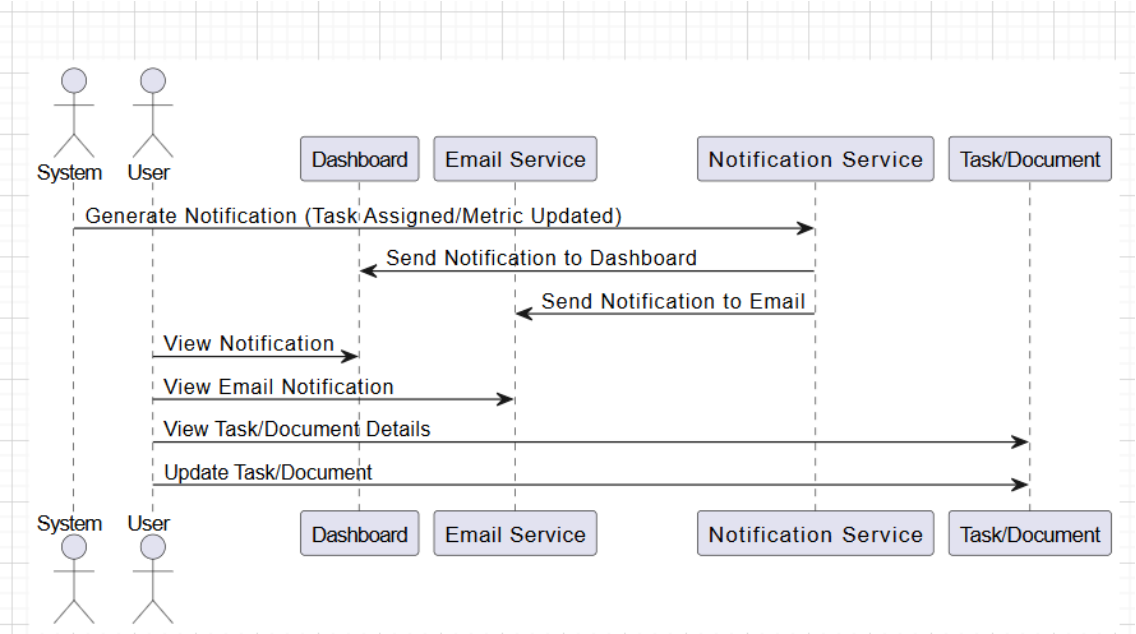

tester
Actor
Test Login and Access Dashboard
Test Fetch and Display KPIs
Test Filter and Customize View
Test Download Reports
System
Actor

5 - Sequence diagrams:

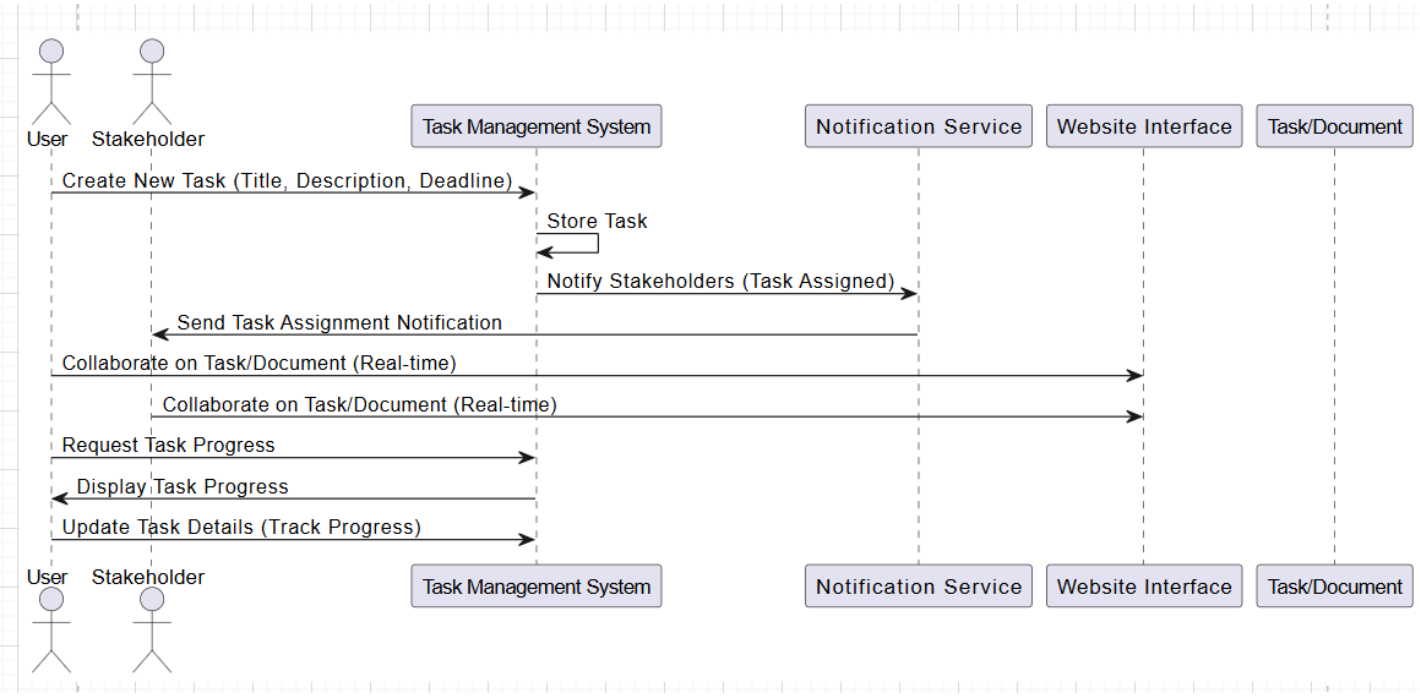
5.1 - First Requirement:



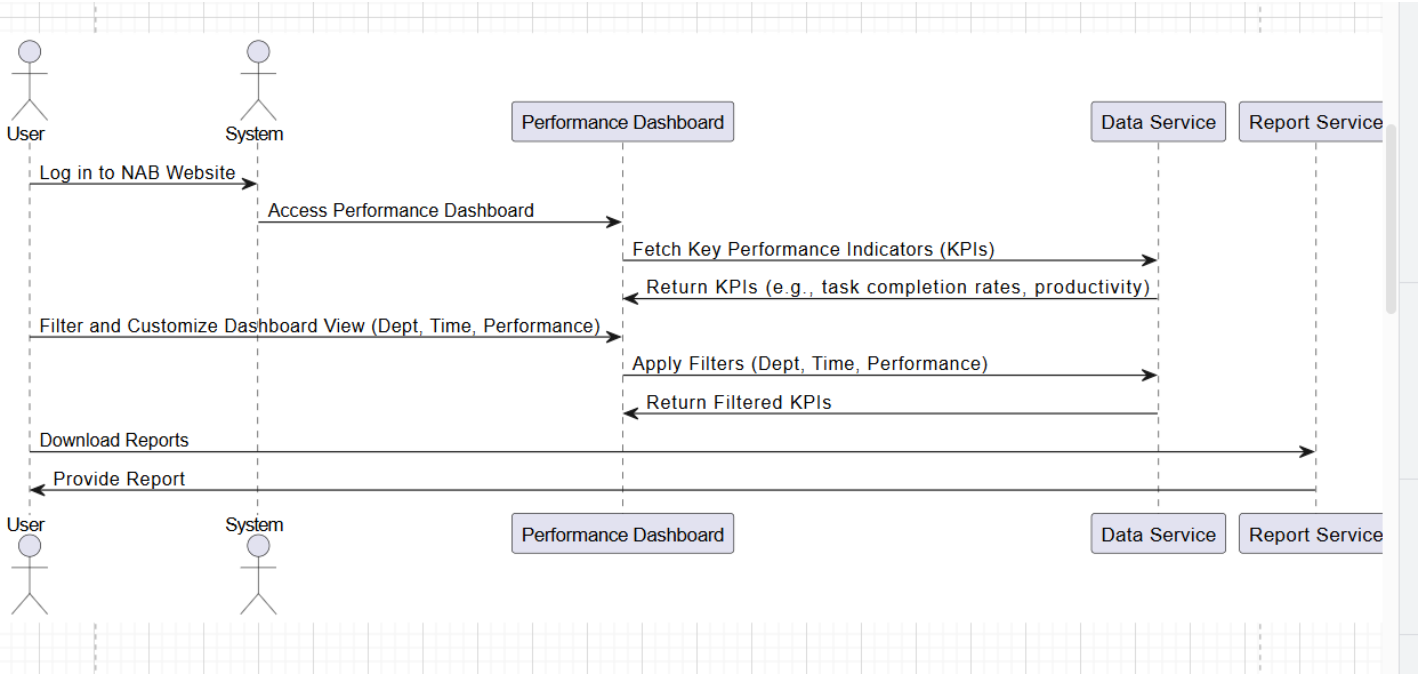
5.2 - Second requirement:



5.3 - Third Requirement:

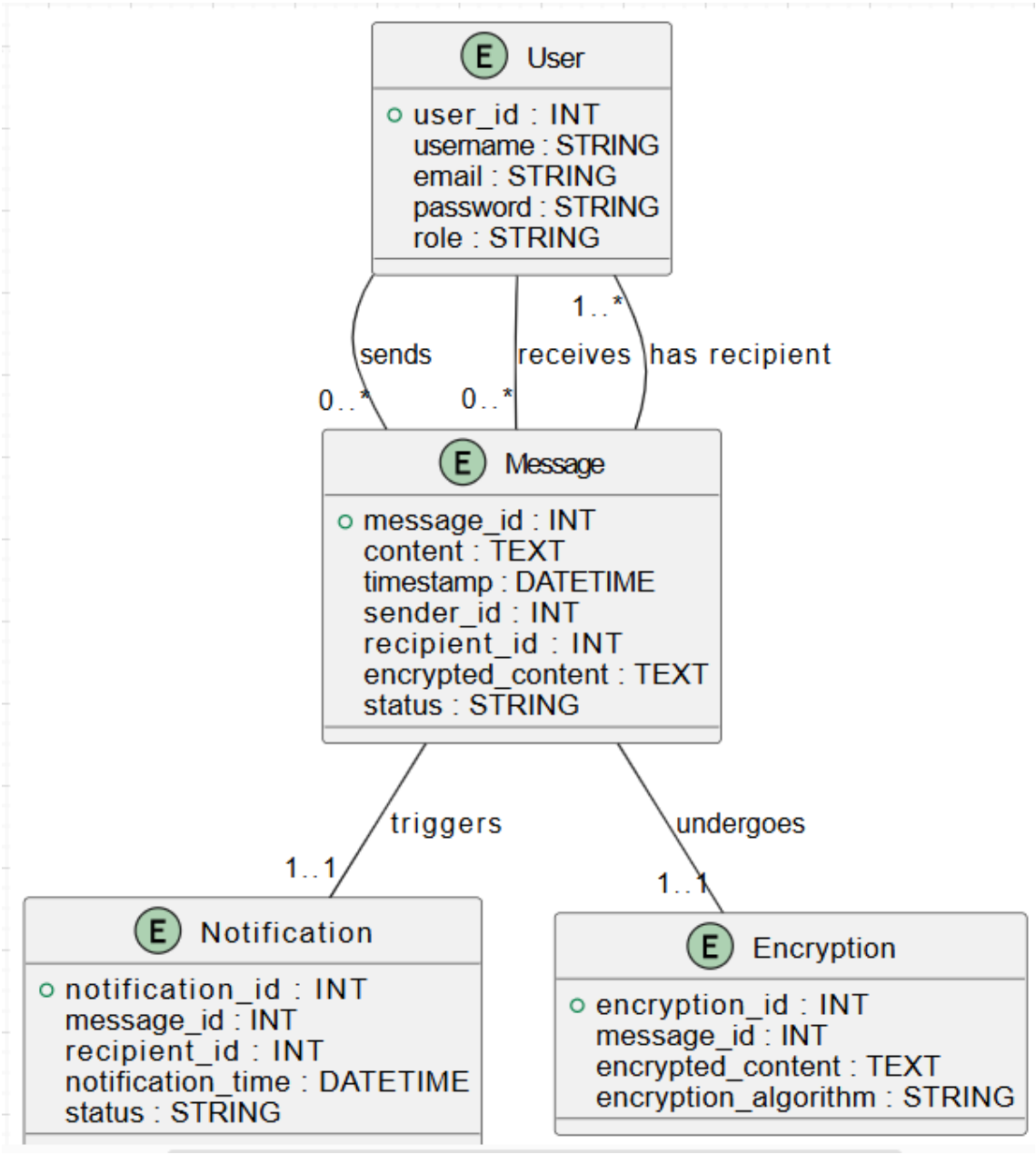


5.4 - Fourth Requirement:

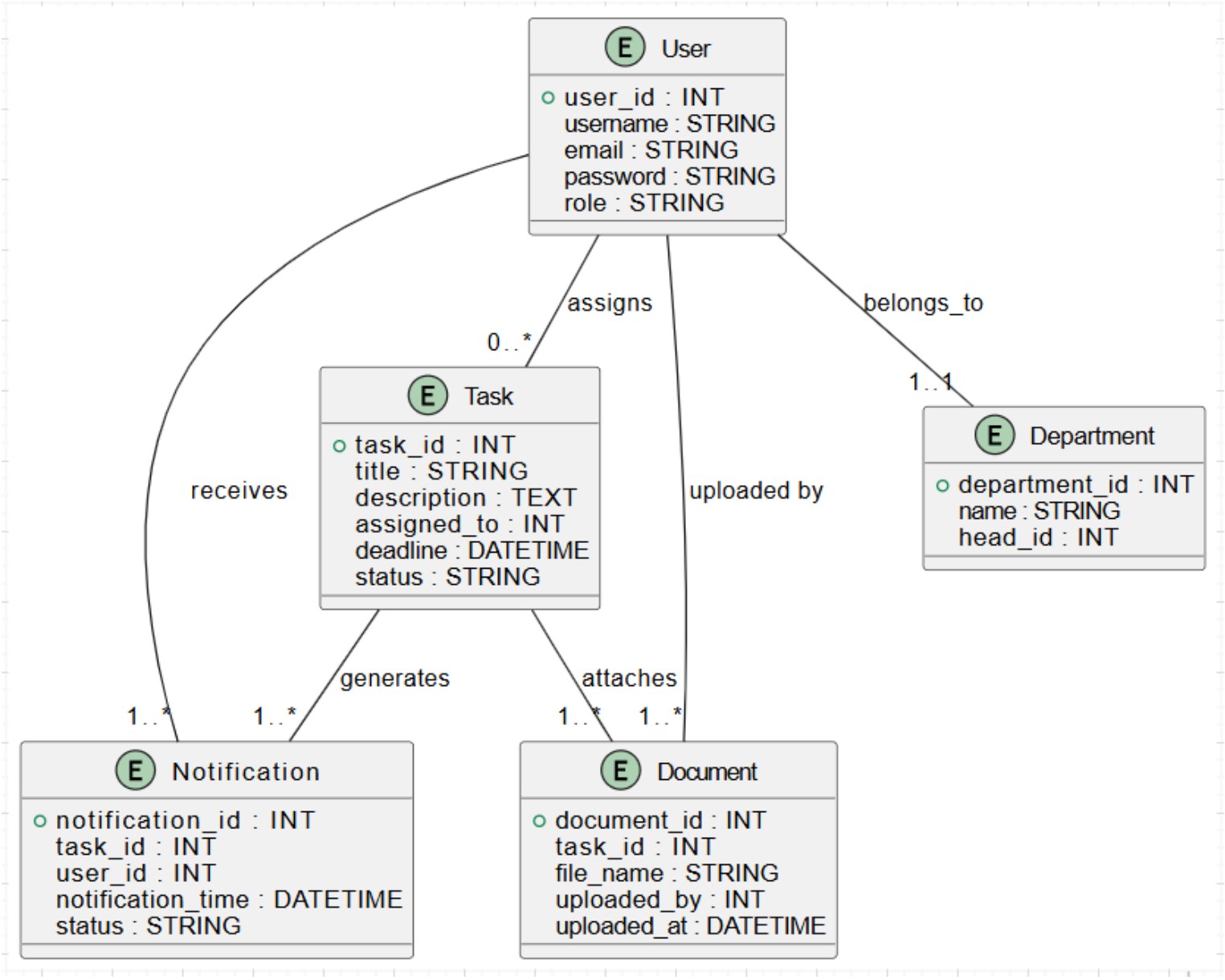


6 - ENTITY-RELATIONSHIP DIAGRAMS:

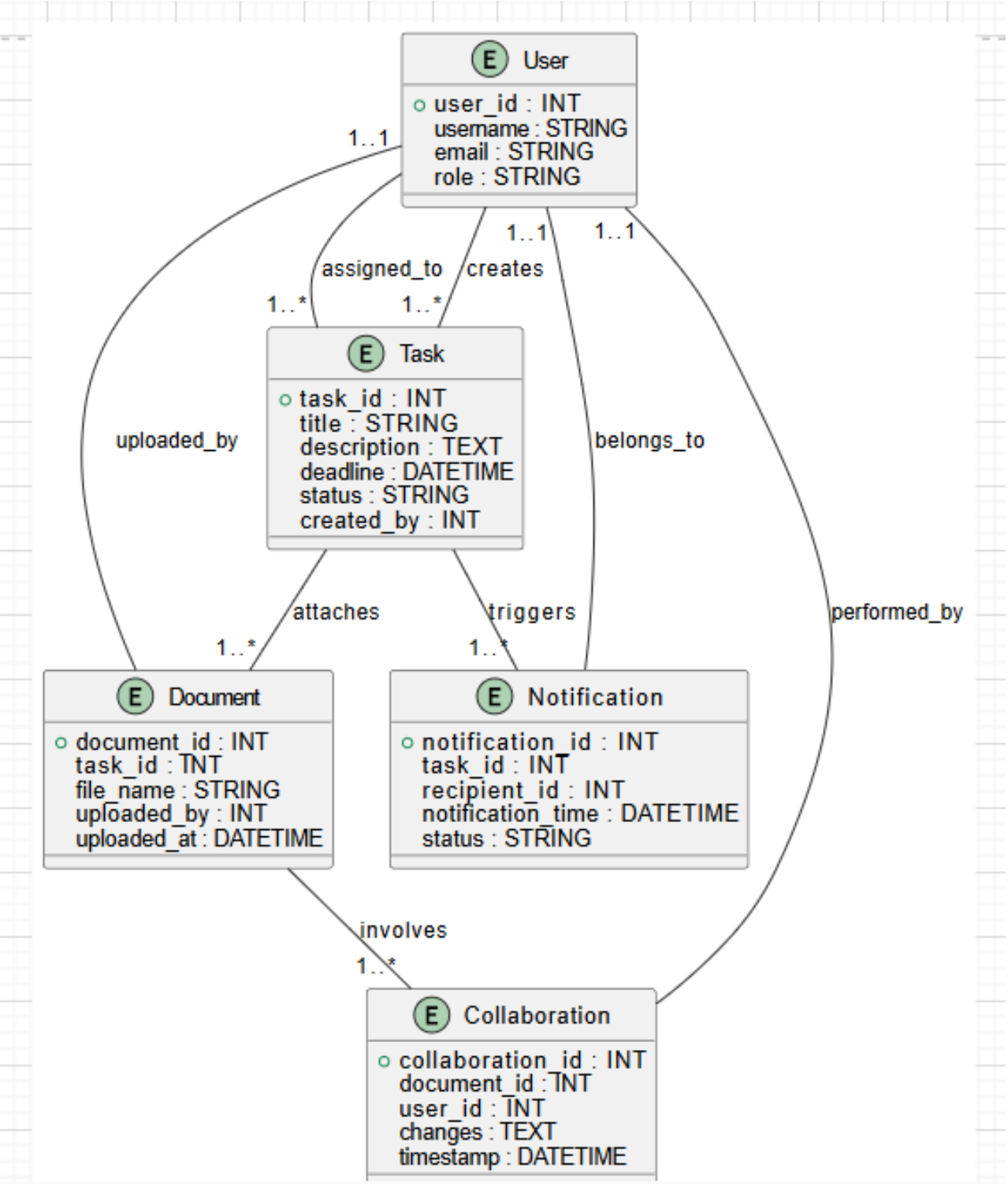
6.1 - First Requirement:



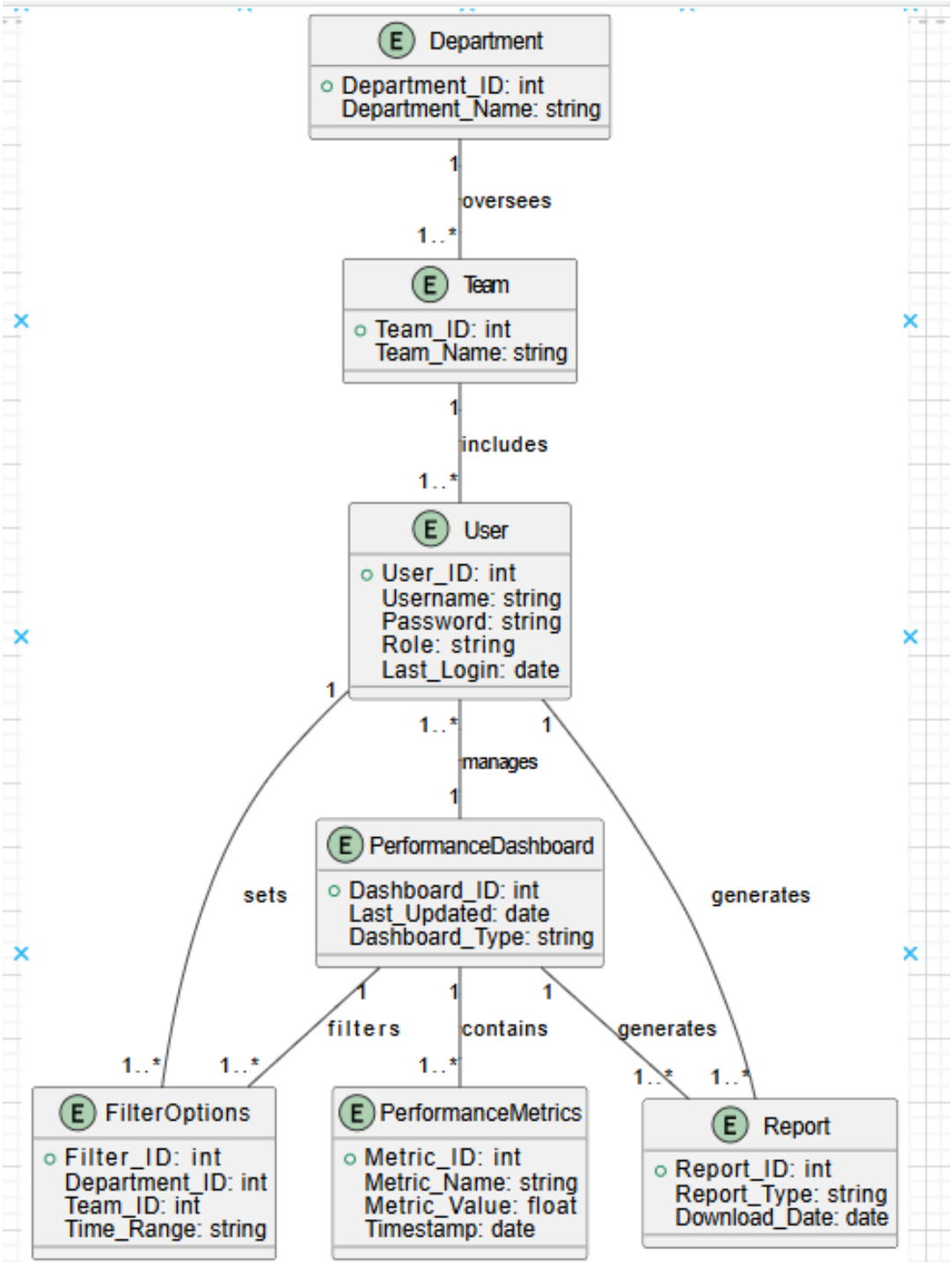
6.2 - Second Requirement:



6.3 - Third Requirement:

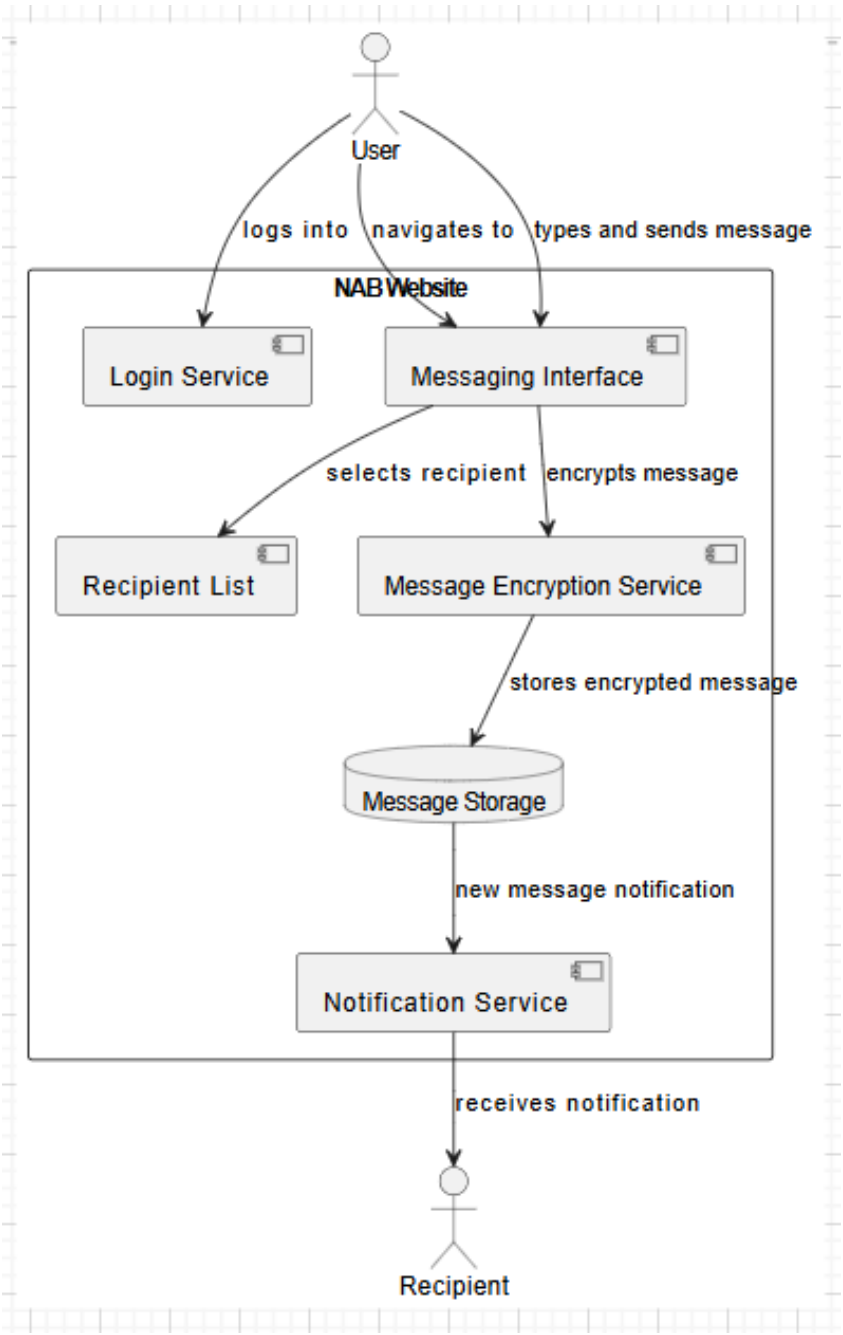


6.4 - Fourth Requirement:

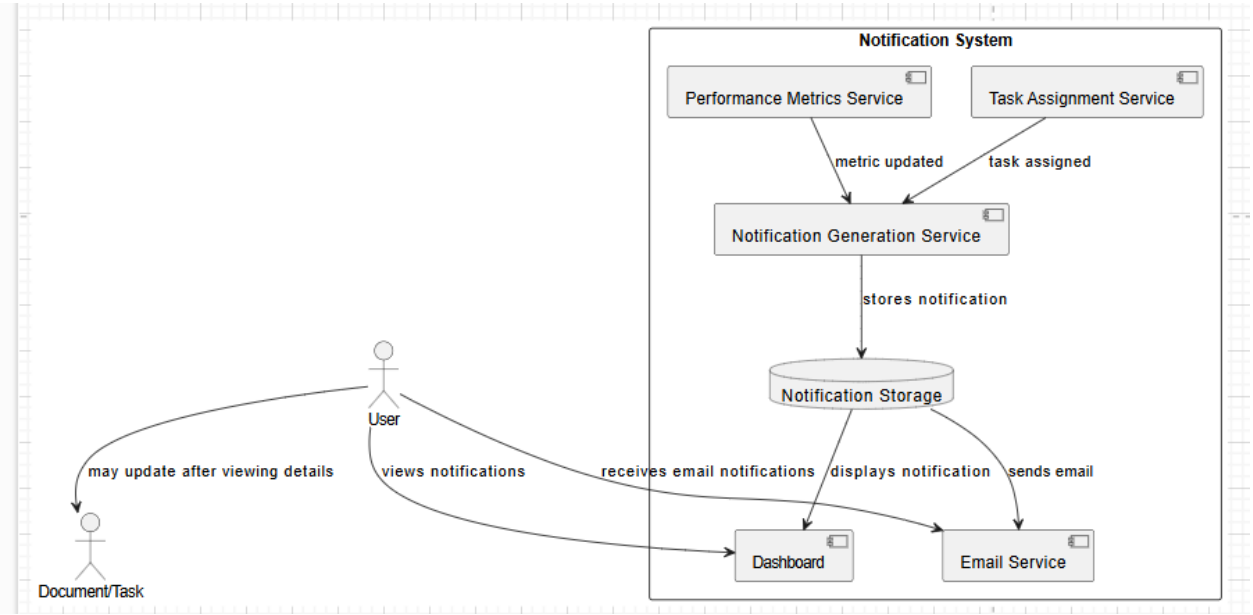


7 - Architecture Design Diagrams:

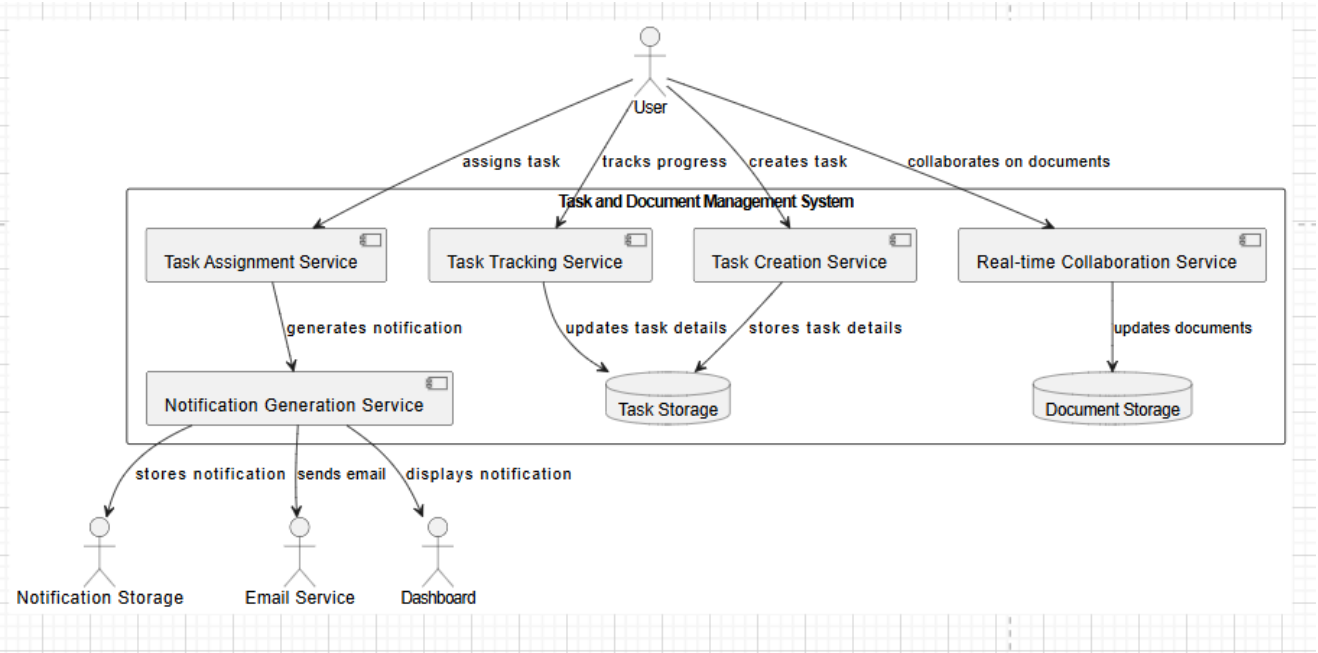
7.1 - First Requirement:



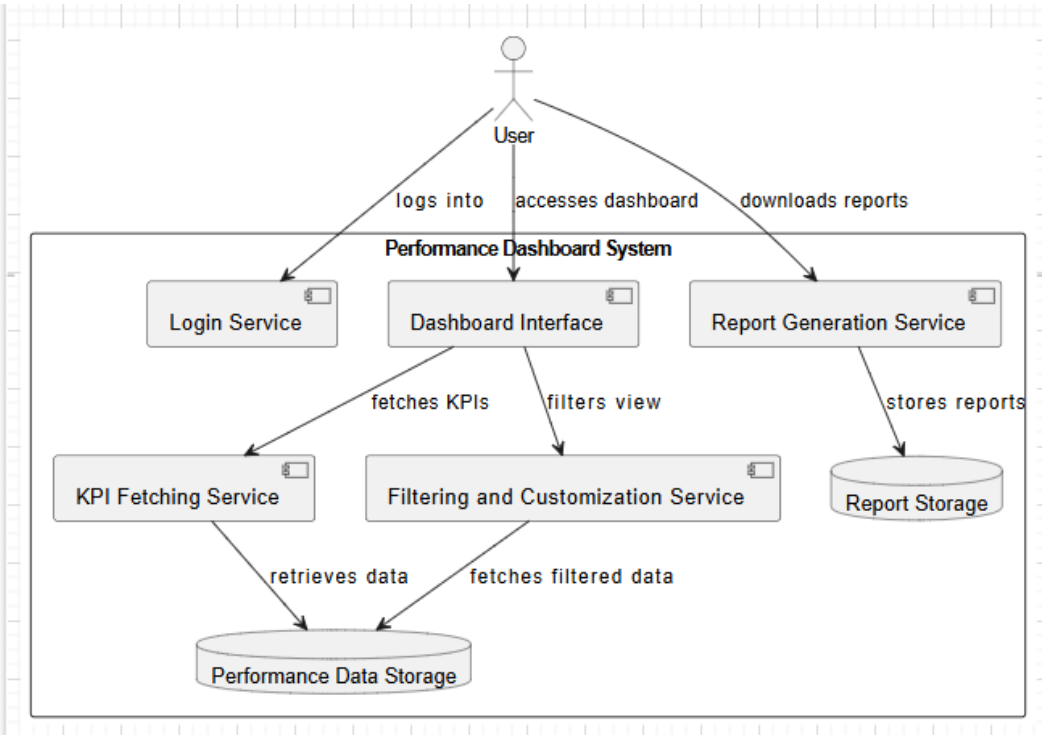
7.2 - Second Requirement:



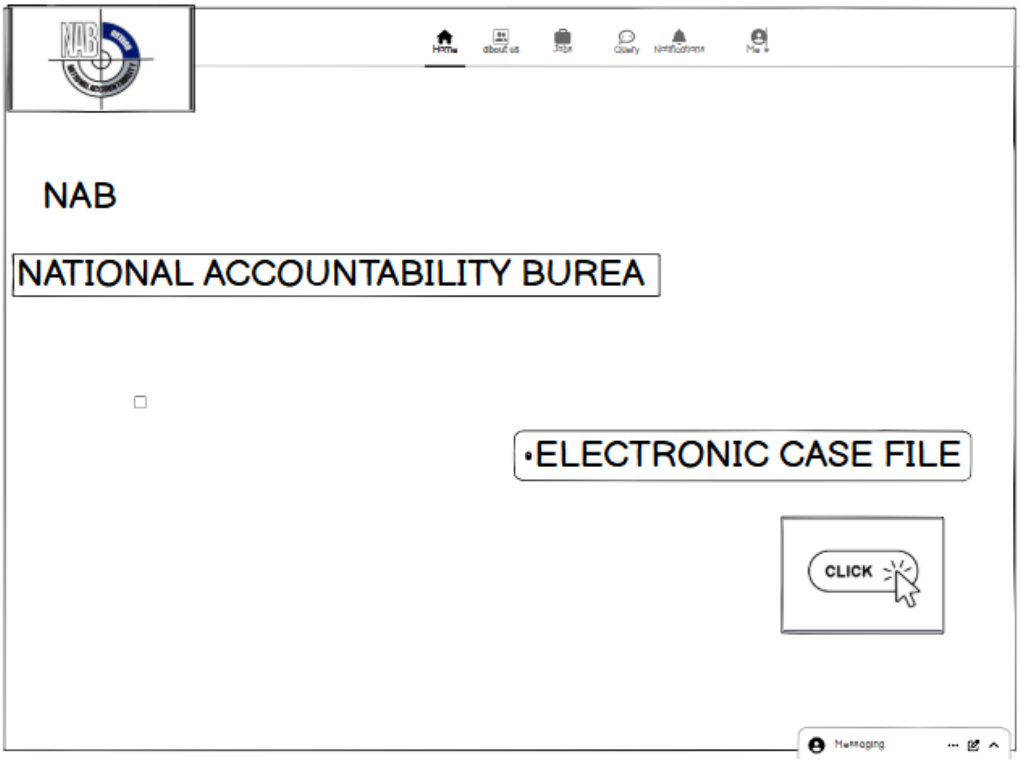
7.3 - Third Requirement:

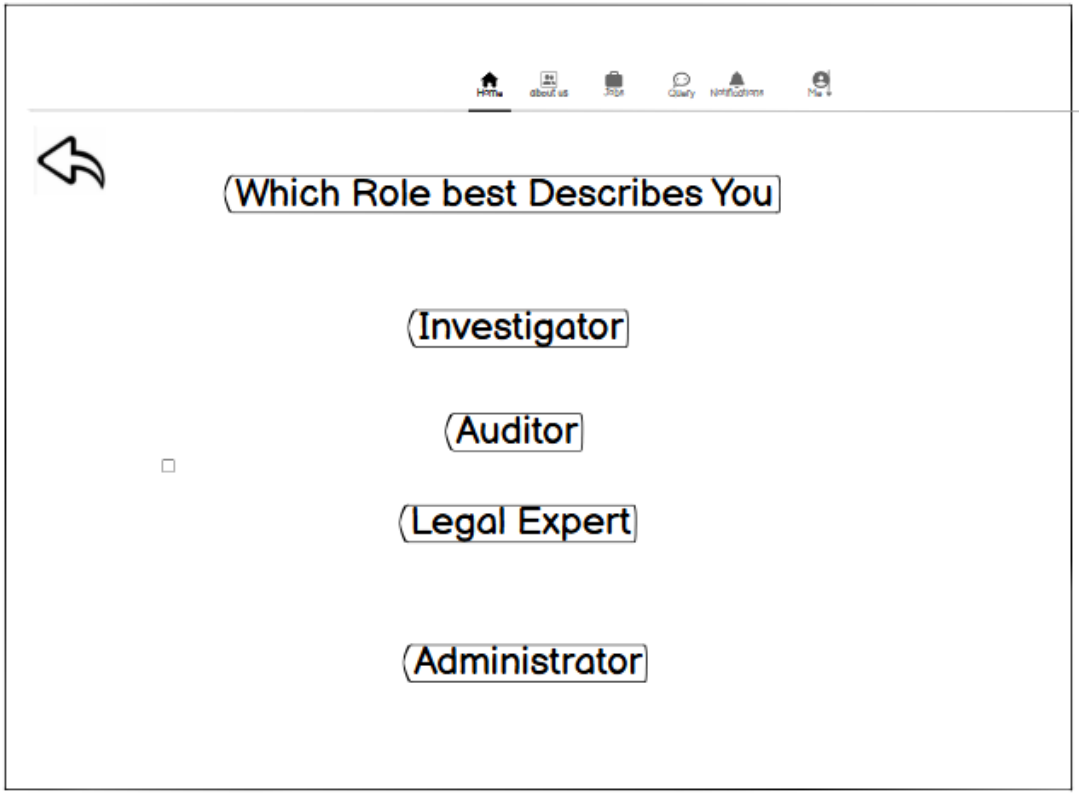


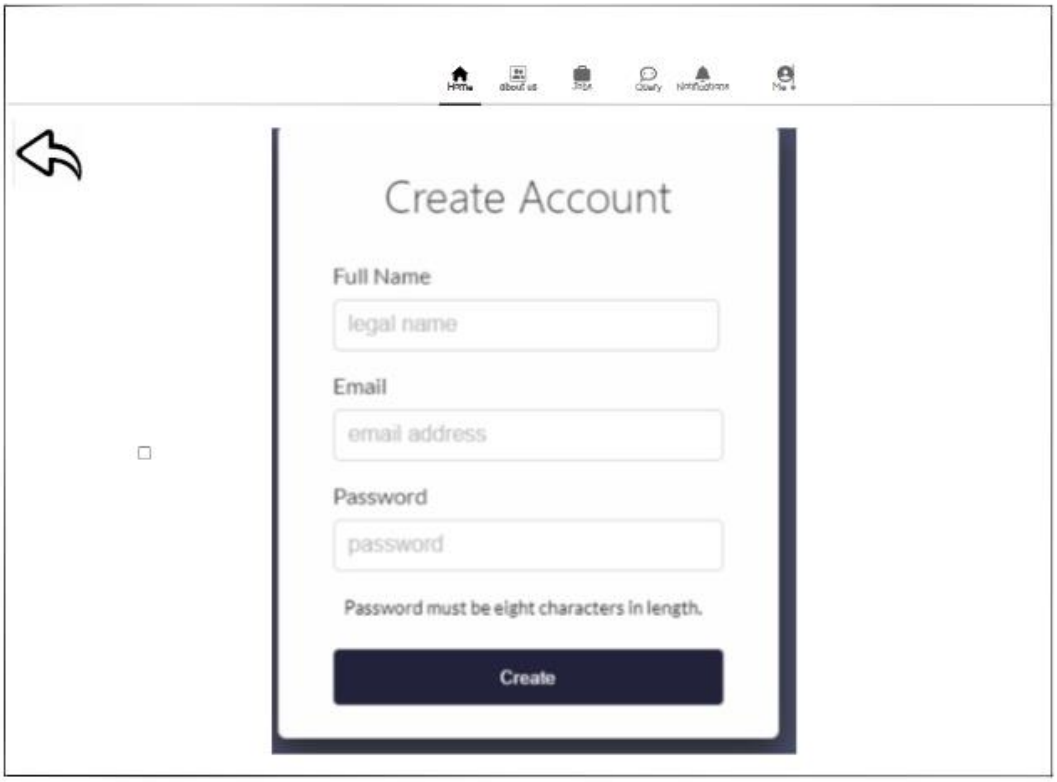
7.4 - Fourth Requirement:



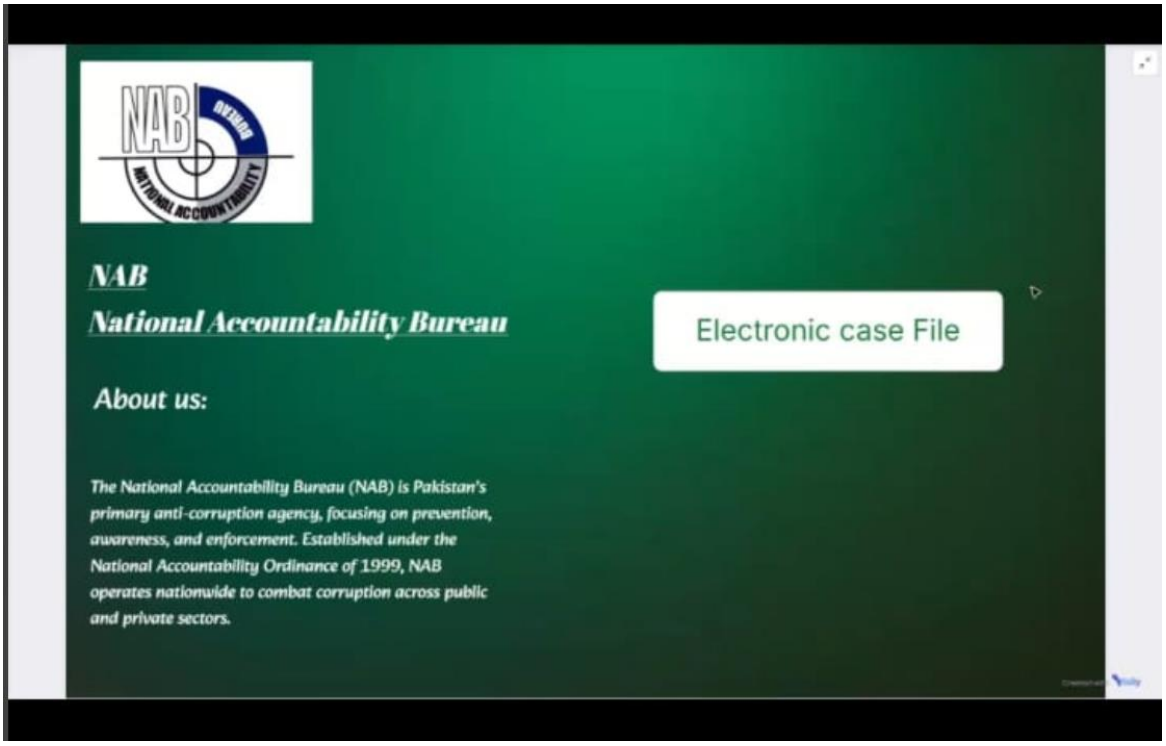
7.5 Wireframe Visualization:

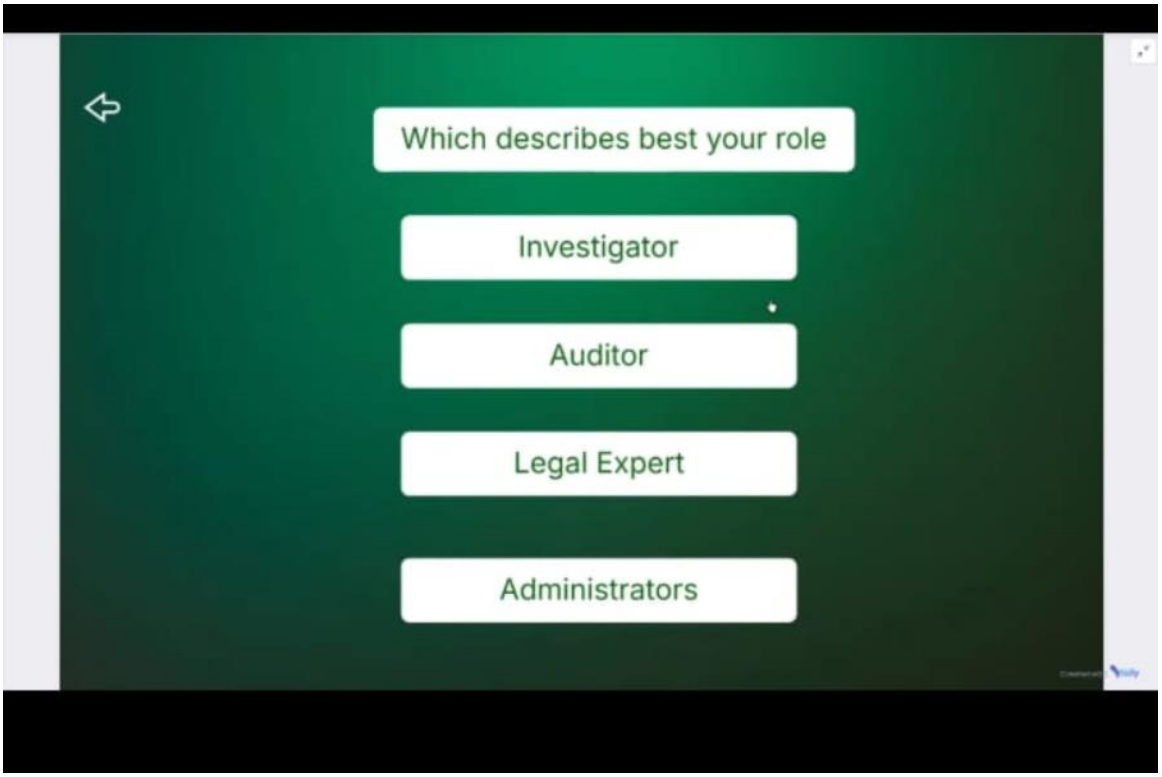



Which Role best Describes You
Investigator
Auditor
Legal Expert
Administrator

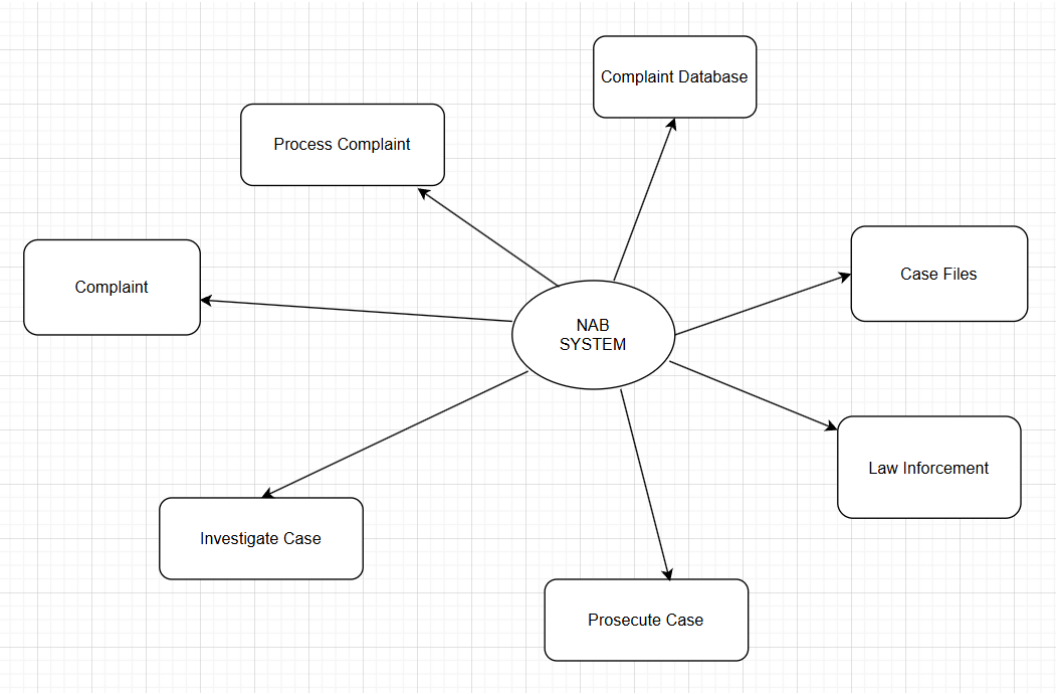

Create Account
Full Name
legal name
Email
email address
Password
password
Password must be eight characters in length.
Create

7.6 - Mockups:


NAB
National Accountability Bureau
Electronic case File
About us:
The National Accountability Bureau (NAB) is Pakistan's primary anti-corruption agency, focusing on prevention, awareness, and enforcement. Established under the National Accountability Ordinance of 1999, NAB operates nationwide to combat corruption across public and private sectors.


Which describes best your role
Investigator
Auditor
Legal Expert
Administrators

7.7 - Data Flow Diagram:



8 - Stakeholder Inputs and Responses:

This part outlines the key inputs gathered from NAB employees and how these have been used in the final SRS.

Stakeholder Meetings:

There were a number of stakeholder meetings with different groups of NAB employees, such as:
- Investigators:
- Administrative Staff:
- IT Department
- Management

Important Inputs and Actions Taken

1. Efficient Case Management:

- Input: Investigators wanted an efficient and smooth case management system.

- Action: The SRS has defined requirements for a centralized case management system with functionalities such as case tracking, document management, and real-time updates

2. User-Friendly Interface:

- Input: The administrative staff needed an interactive interface to input and manage their data.
- Action: SRS has requirements that have a user-friendly interface, interactive and well-coordinated with proper instructions, having enough support options.

3. Stronger Security Measures:

- Input: The IT department considered the proper handling of the data with extreme security measures so that the confidential information was not misused.
- Action: SRS has ensured specifications for the improved security details like encryption, safe authentications, and regular security checkups.

4. Integration with existing systems:

- Input: The IT department felt it should seamlessly fit into the NAB systems and databases already in place.
- Action: The SRS defines all the requirements for API-based integration with all the current systems and synchronization of data with all of their current systems.

5. Real-time Collaboration:

- Input: what employees needed was support toward the sharing and utilization of real-time data to foster teamwork.
- Action: Needs for collaborating tools securely integrated into the system natively in the SRS.

9 - Traceability Matrix

REQUIREMENT	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14
R1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R2	0	0	0	0	0	0	0	0	0	1000	0	0	0	0
R3	0	0	0	0	0	0	1000	1000	1	0	0	0	0	0
R4	0	0	0	0	0	0	0	0	0	0	1000	0	0	0
R5	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R6	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R7	0	0	1000	0	0	0	0	1000	1	0	0	0	0	0
R8	0	0	1000	0	0	0	1000	0	1	0	0	0	0	0
R9	0	0	1000	0	0	0	1	1	0	0	0	0	0	0
R10	0	1000	0	0	0	0	0	0	0	0	0	0	0	0
R11	0	0	0	1000	0	0	0	0	0	0	0	0	0	0
R12	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R13	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R14	0	0	0	0	0	0	0	0	0	0	0	0	0	0